

Can We Predict Apartment Prices in Vitoria Based on Apartment Attributes?

Introduction

Data on apartment attributes and market prices in Vitoria, Spain have been obtained from the VIT2005 dataset. This report aims to investigate the following:

- Can we predict the apartment prices in Vitoria based on various apartment attributes?

Exploratory Analysis

Figure 1 below shows the paired scatterplots of all variables. It suggests linear relationships between total price and all variables, except for age, where the relationship is less clear. Therefore, by the first impression, we consider the inclusion of all the predictor variables in the model.

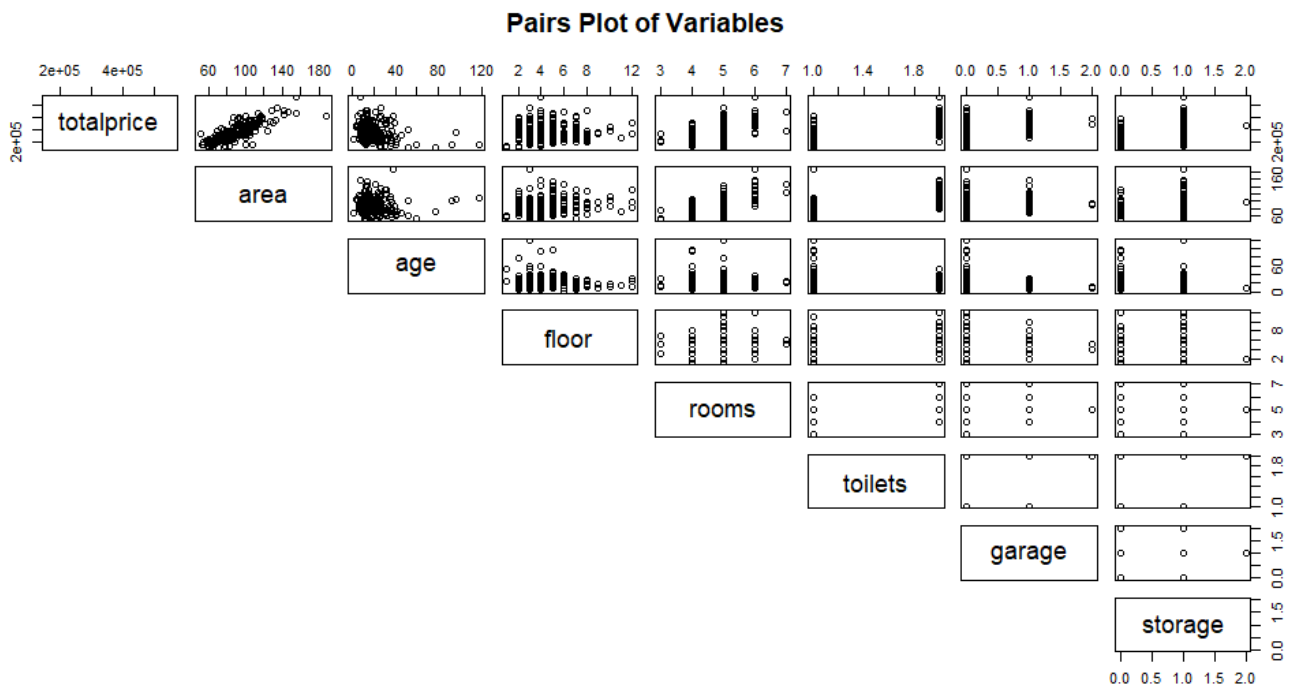


Figure 1: Pairs plot of all variables

Statistical Analysis

To build a model, we perform a criterion-based variable selection by maximising R_{adj}^2 . By the R-output, we find the best model selects 6 predictor variables (area, age, floor, rooms, toilets, garage, and storage) giving the maximum $R_{adj}^2 = 0.7655030$. Figure 3 further illustrates the variables included in best model from 1 to 7 variables.

```
Subset selection object
Call: regsubsets.formula(totalprice ~ ., data = df)
7 Variables (and intercept)
Forced in Forced out
area      FALSE    FALSE
age       FALSE    FALSE
floor     FALSE    FALSE
rooms     FALSE    FALSE
toilets   FALSE    FALSE
garage    FALSE    FALSE
storage   FALSE    FALSE
1 subsets of each size up to 7
Selection Algorithm: exhaustive
   area age floor rooms toilets garage storage
1 ( 1 ) ** *  " " " " " " " " " " " "
2 ( 1 ) ** *  " " " " " " " " " * " "
3 ( 1 ) ** *  " * " " " " " " " " " "
4 ( 1 ) ** *  " * " " " " " " " * " "
5 ( 1 ) ** *  " * " " " " " " " * " "
6 ( 1 ) ** *  " * " " * " " " " * " "
7 ( 1 ) ** *  " * " " * " " * " " * " "
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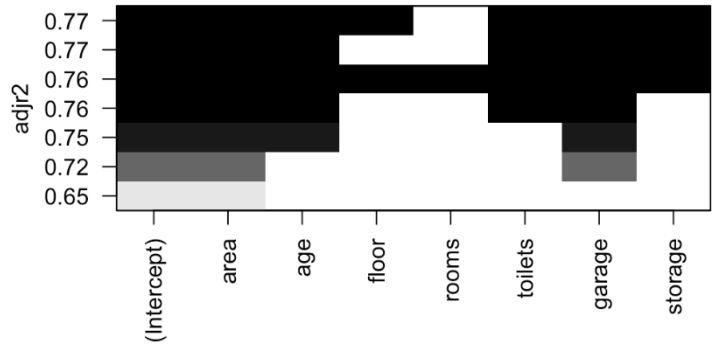


Figure 2: R output of the model selection

Figure 3: Plot of variables by against R_{adj}^2

No of Variables	1	2	3	4	5	6	7
R_{adj}^2	0.6532269	0.7175451	0.7494421	0.7615749	0.7653295	0.7655030	0.7645447

Table 1: R_{adj}^2 of selected models

Therefore, our model which includes 6 variables is

$$\mathbb{E}(\text{Totalprice}) = \beta_0 + \beta_1(\text{area}) + \beta_2(\text{age}) + \beta_3(\text{floor}) + \beta_4(\text{rooms}) + \beta_5(\text{toilets}) + \beta_6(\text{garage}) + \beta_7(\text{storage})$$

By Figure 4 we find $\hat{\beta}_i \forall i$, and so the final model is

```

Call:
lm(formula = totalprice ~ area + age + floor + toilets + garage +
    storage, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-91086 -18875  -1866   16057 101506

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  64228.5    12707.0   5.055 9.34e-07 ***
area          2104.0     144.1   14.598 < 2e-16 ***
age          -664.2     171.2   -3.881 0.000139 ***
floor        -1181.9    1098.8   -1.076 0.283348
toilets       21881.8    6441.3   3.397 0.000814 ***
garage        29105.6    5767.2   5.047 9.69e-07 ***
storage       11338.8    5974.8   1.898 0.059090 .
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33560 on 211 degrees of freedom
Multiple R-squared:  0.772,    Adjusted R-squared:  0.7655
F-statistic: 119.1 on 6 and 211 DF,  p-value: < 2.2e-16

```

Figure 4: Summary of the model

$$\mathbb{E}(\text{Totalprice}) = 64228.5 + 2104(\text{area}) - 644.2(\text{age}) - 1181.9(\text{floor}) + 21881.8(\text{toilets}) + 29105.6(\text{garage}) + 11338.8(\text{storage})$$

To interpret our final model, the Total price increases by €2104.0 as area increases by 1m² with all other predictor variables held constant, and the interpretations of coefficients for the other covariates follow in an equivalent way. Note that there is no practical interpretation of the intercept in this model because an apartment with predictor variables being 0 (e.g. area) is unrealistic. The interpretation of our R_{adj}^2 value is that our model explains 76.55% of the variance in the total price with penalisation of additional covariates, which is considered a high value and thus justifies the appropriateness of our model.

In the exploratory analysis we noted that the relationship between age and total price was less clear compared to the other predictor variables. Therefore, we apply 4 transformations to try to better harmonise assumptions of linear regression and perform a model selection as above to see if the

R^2_{adj} of these models is higher. By Table 2 below, we see that our final model with no transformations remains the best based on R^2_{adj} criterion, which further justifies the appropriateness of our model.

Transformation	None	log(age)	1/age	sqrt(age)	log(totalprice)
R^2_{adj}	0.7655	0.7562	0.7506	0.7607	0.7650

Table 2: Adjusted R-squared after various transformations

We now check the assumptions for our model. Figure 5 given below shows the errors randomly scatter around zero versus fitted values, which justifies the assumptions of the mean error being 0 and constant variability. Figure 5 also shows the errors randomly scatter versus the errors observed previously, which justifies the assumption of independence of errors. By Figure 6, we have a reservation about the normality assumption because the residuals deviate from the theoretical normal at the tails, but we assume it holds for model interpretation. Figure 7 below shows residuals vs the predictor variables randomly scatter around 0, which justifies the assumption of linear relationship.

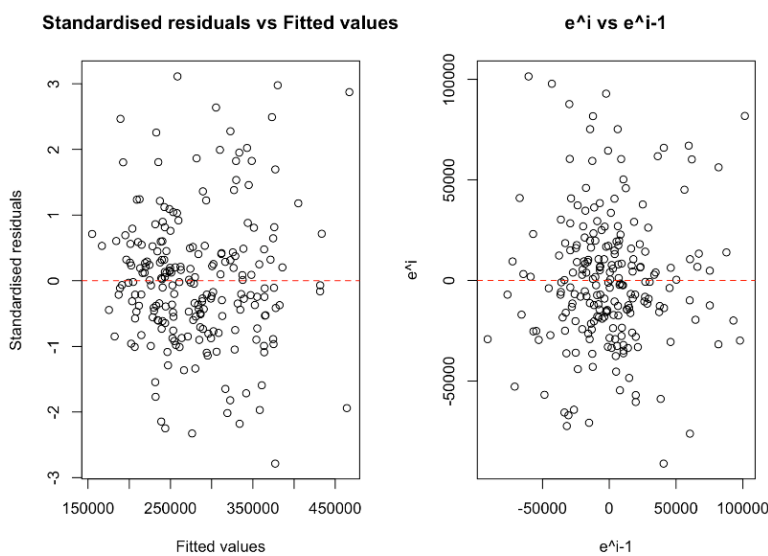


Figure 5: Scatterplots

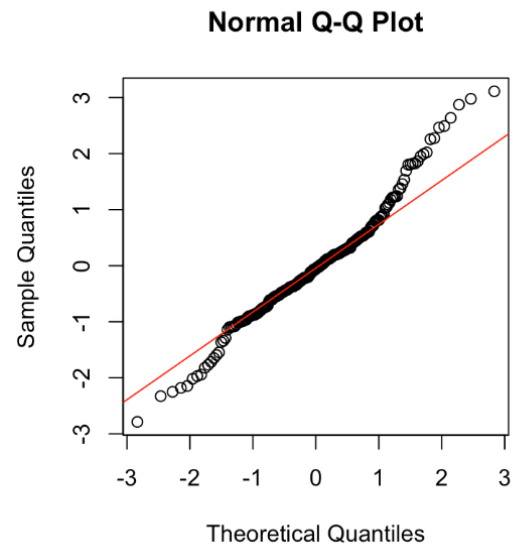


Figure 6: QQ-plot of standardised residuals

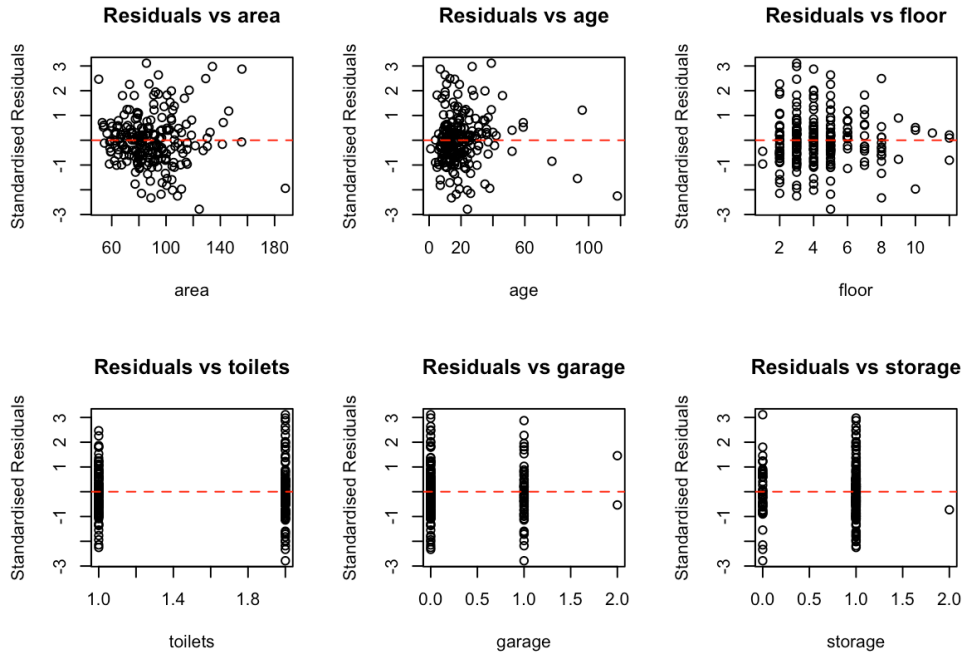


Figure 7: Residuals vs the predictor variables

We now make a prediction for an apartment with area = 100, age = 50, floor = 3, toilets = 1, garage = 1, storage = 1. By Table 3 below, we ensure we are making a sensible prediction and Table 4 shows the confidence and prediction interval for the observation. Our interpretation of the point estimate is for an apartment with the specified attributes, the expected Total price is €300194.4. Meanwhile for the confidence interval and prediction interval, our interpretation is that we are 95% confident that the population mean Total price for an apartment with the specified attributes lies in the interval [€283591.9, €316797], and that we are 95% confident that a future apartment's price with the specified attributes lies in the interval [€231991.5, €368391.4] respectively, assuming the normality assumption holds.

Variable	area	age	floor	toilets	garage	storage
Range	[50.38,187.91]	[1,118]	[1,12]	[1,2]	[0,2]	[0,2]

Table 3: range of predictor variables in the data

Point estimate	300194.4
95% Confidence Interval	[283591.9, 316797]
95% Prediction Interval	[231991.5, 368397.4]

Table 4: R-output of point estimate, confidence interval, and prediction interval

Conclusion

The analysis indicates that apartment prices in Vitoria can be reasonably predicted using property characteristics.

In our final model, the predictor variable rooms was not included, despite appearing to have a strong linear relationship with total price in exploratory analysis. This could be due to multicollinearity between rooms and toilets (which was included in the model), as number of rooms is clearly correlated with number of toilets.

Revisiting our reservation about the normality assumption, the consequence is that our confidence and prediction intervals may no longer be reliable. Therefore, further analysis should be conducted to verify the normality assumption, perhaps through a formal hypothesis test.

A further analysis of the data could include different criterion based approaches (e.g. Mallow's C_p or BIC), or stepwise approaches. This may result in different models, due to correlation between predictor variables.